Appendix 5: Improvements and Refinements to the Blockchain-Based Design Pattern Decision Model

This document focuses on the improvements made to a decision model for blockchain-based design patterns, incorporating expert feedback to address previous limitations and enhance its practicality. Key updates include the addition of new categories, design patterns, and refined quality attribute mappings to ensure a more comprehensive framework. The model has been improved to better align with real-world blockchain needs, offering clearer structure, expanded scope, and enhanced usability.

Specifically, major revisions include the addition of missing design patterns, such as Tokenisation and Incentive Execution, and the refinement of quality attribute mappings with clearer trade-offs, like transparency and reliability. Inherent attributes, such as optimizing traditional finance, were added for contextual clarity, and unique identifiers were introduced for easier navigation. New categories, including the Incentive Layer and Contract Migration, address critical areas like economic mechanisms and cross-chain operability. These updates ensure the model provides actionable and detailed guidance for blockchain design. The following sections outline these revisions and compare the original and updated models.

1 Decision Model for Blockchain-Based Design Patterns

L1: Data Layer

C1.1: Data Interaction

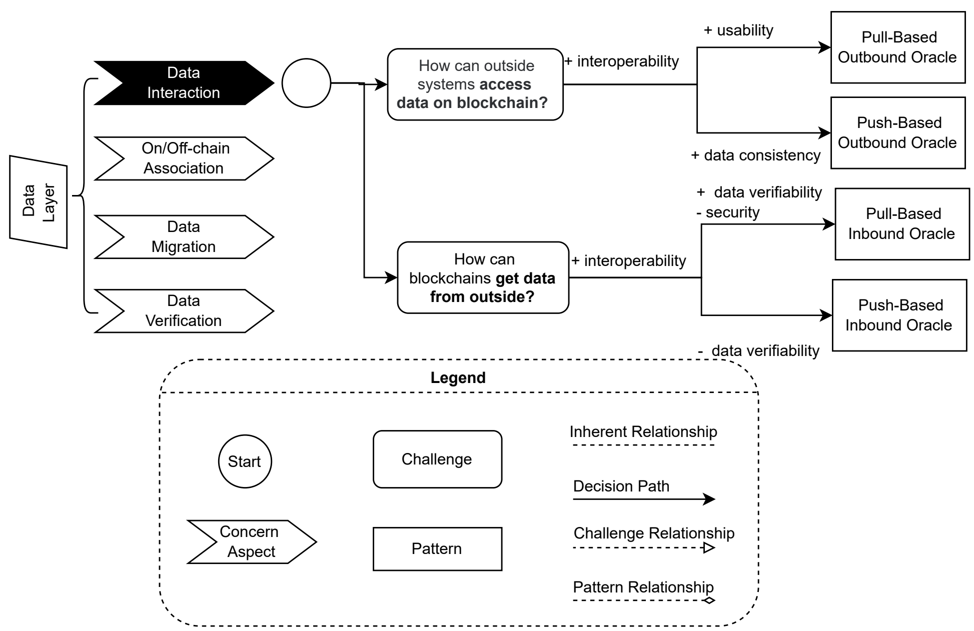

Fig. 1: Original Version: Decision Model for Data Interaction (C1.1)

Updates to the Decision Model: The Data Interaction model was refined to address previously overlooked challenges related to data validation and consistency. Updates include the addition of new design patterns and more precise mappings to quality attributes. The updates are summarized as follows:

1) **Addition of Missing Design Patterns:** Patterns such as *Select-Storage* and *Bulletin Board* were added to ensure comprehensive coverage.
2) **Inclusion of Inherent Attributes:** Inherent attributes like *Rely on off-chain components* were introduced to address previously overlooked considerations.
3) **Six-Layer Blockchain Architecture Integration:** The model now incorporates the six-layer blockchain architecture, as shown on the left side of the diagram, highlighting the position of the current layer within the overall framework.

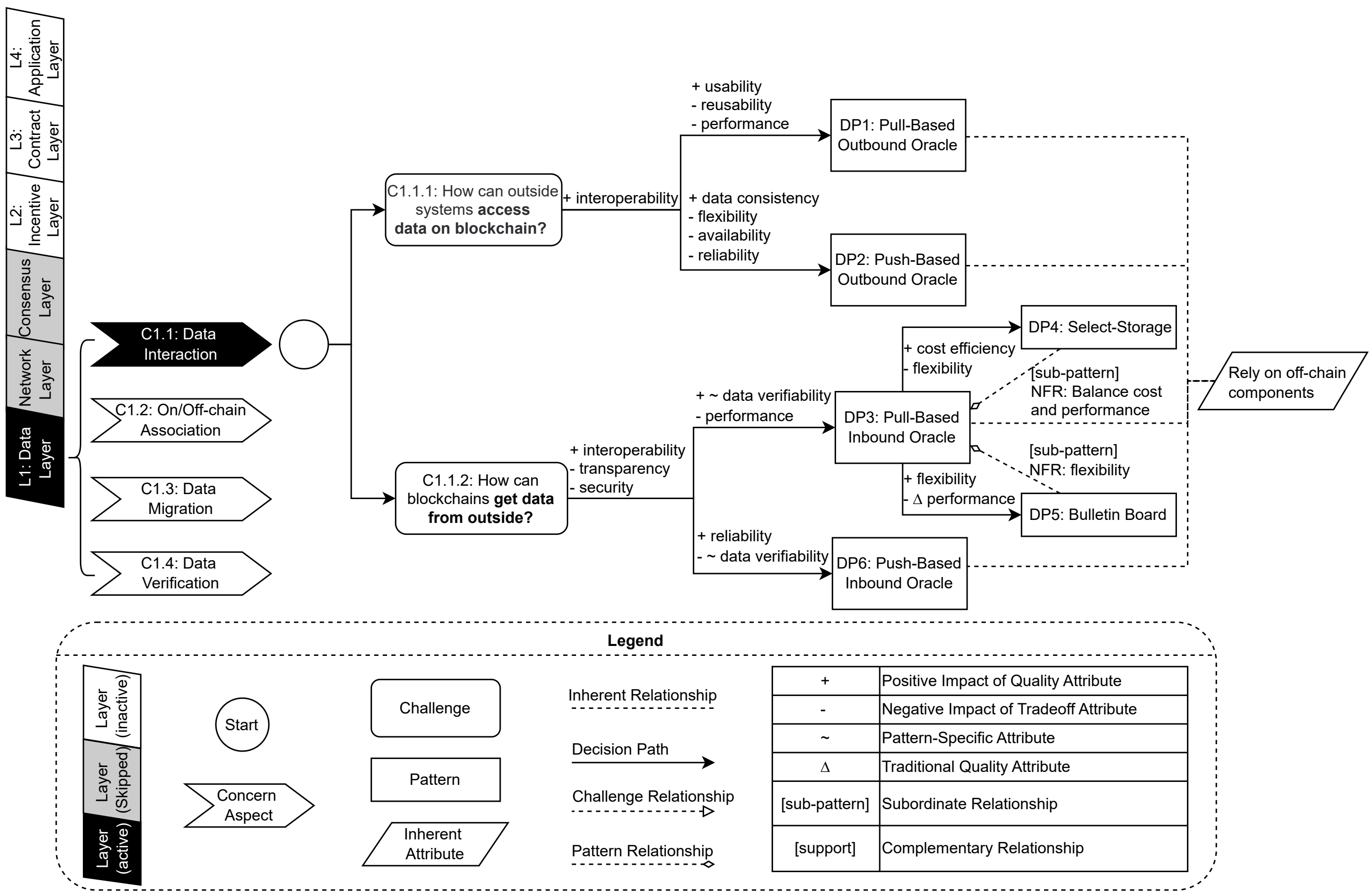

Fig. 2: Updated Version: Decision Model for Data Interaction (C1.1)

4) **Defined Relationships Between Design Patterns:** Relationships between patterns are now explicitly shown, highlighting dependencies and interactions. For example, the *Pull-Based Outbound Oracle* pattern supports *Push-Based Outbound Oracle* by balancing on-chain cost and performance, while *Bulletin Board* focus on providing better flexibility.
5) **Enhanced Quality Attribute Scope:** Symbols such as Δ and $\sim$ were added to indicate trade-offs and impacts of quality attributes.
6) **Expanded Legend:** The legend now includes all symbols and relationships for easier interpretation.
7) **Improved Attribute Mapping:** The original model lacked discussions on certain quality attributes, such as reusability and flexibility. The updated version includes these attributes which strengthen decision-making outcomes.
8) **Item Numbering:** Unique identifiers were assigned to all items in the model for easier navigation and reference. including concerns to address (*C1.1, C1.2, C1.3, C1.4*), challenges (*C1.1.1, C1.1.2*), and patterns (*DP1, DP2, DP3, DP4, DP5, DP6*) were labeled systematically, ensuring consistency and facilitating quick identification of specific components within the decision model.

C1.2: On/Off-Chain Association

Updates to the Decision Model: Improvements to the On/Off-Chain Association model focused on refining the mapping of quality attributes such as scalability and data verifiability, as well as adding new elements to better describe relationship between challenges. The updates are summarized as follows:

1) **Addition of Missing Design Patterns:** The pattern *DP9: Content-Addressable Storage* was added to cover previously overlooked solutions relevant to off-chain data management and cost reduction.
2) **Inclusion of Inherent Attributes:** Attributes such as *Rely on off-chain components* were introduced to highlight the dependence of certain solutions on off-chain elements.
3) **Six-Layer Blockchain Architecture Perspective:** The model now aligns with the six-layer blockchain architecture for better context and structure.
4) **Defined Relationships Between Design Challenges:** Relationships between challenges are explicitly defined. For instance, *C1.2.1* (*How to reduce both storage and computation costs on chain?*) supports *C1.2.2* (*How to reduce storage costs on chain?*) through off-chain storage and *C1.2.3* (*How to reduce computation costs on chain?*) through off-chain computation, illustrating how different challenges are interconnected and mutually influence each other.

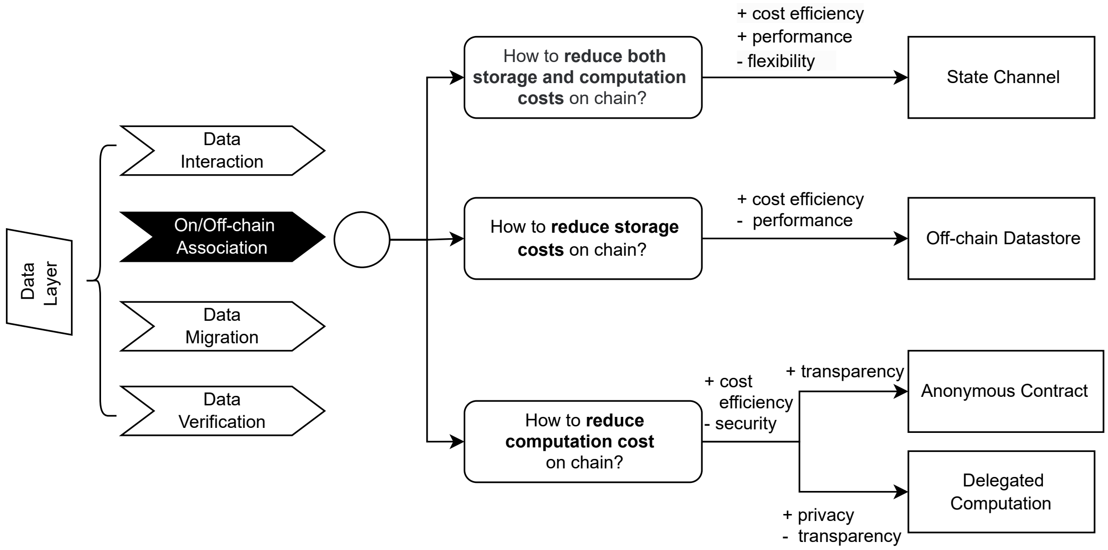

Fig. 3: Original Version: Decision Model for On/Off-Chain Association (C1.2)

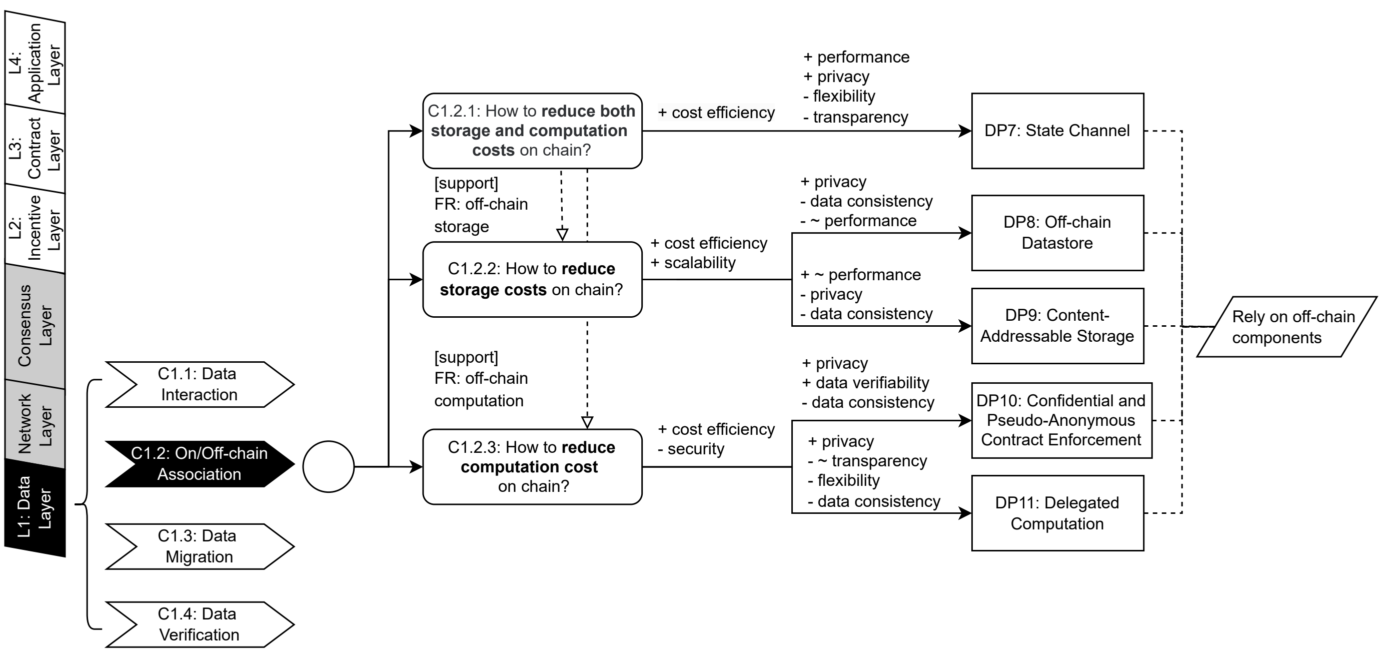

Fig. 4: Updated Version: Decision Model for On/Off-Chain Association (C1.2)

5) **Enhanced Quality Attribute Scope:** Symbols such as Δ and $\sim$ were added to represent the trade-offs and specific impacts of quality attributes, including discussions on performance and transparency.
6) **Improved Quality Attribute Mapping:** Additional attributes like *scalability* and *data verifiability* were mapped to relevant design patterns, ensuring a more comprehensive decision-making process.
7) **Item Numbering:** Unique identifiers were assigned to all items in the model, including concerns to address(*C1.1, C1.2, C1.3, C1.4*) challenges (*C1.2.1, C1.2.2, C1.2.3*) and patterns (*DP7, DP8, DP9, DP10, DP11*), enabling quicker navigation and reference.

C1.3: Data Migration

Updates to the Decision Model: The Data Migration decision model has been revised to address key limitations including addition of inherent attributes and quality attribute mappings, improving its clarity and effectiveness. Below is a list of the major updates:

1) **Addition of Inherent Attributes:** Inherent attributes such as *Only work on arbitrarily created tokens* and *Specific*

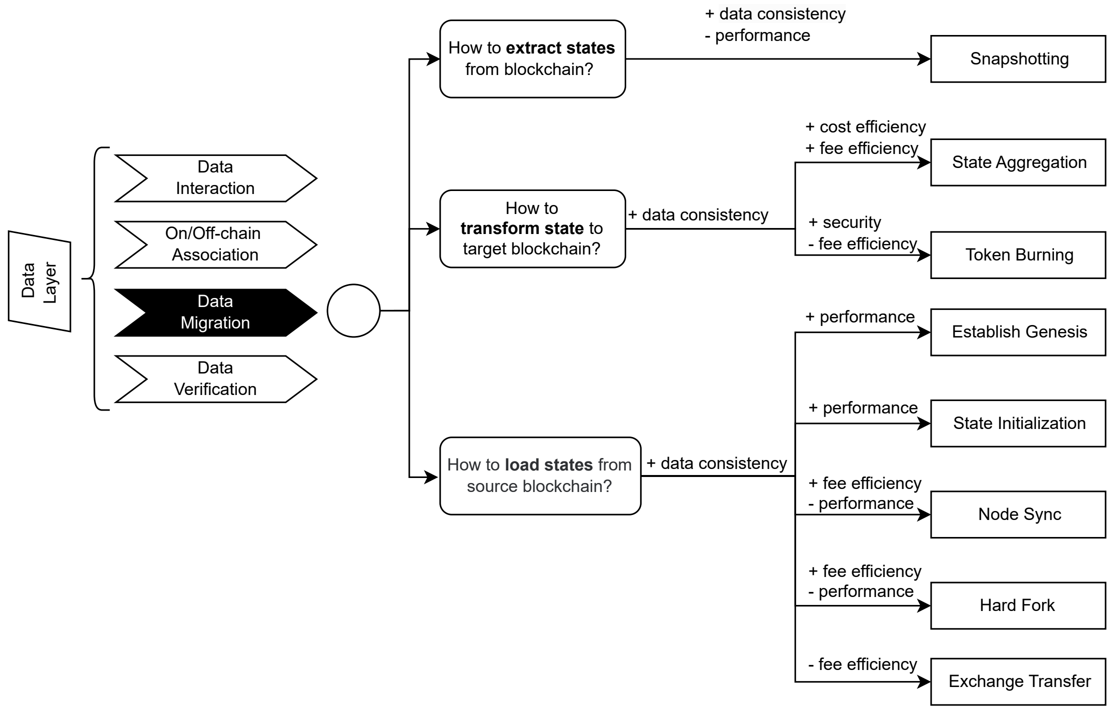

Fig. 5: Original Version: Decision Model for Data Migration (C1.3)

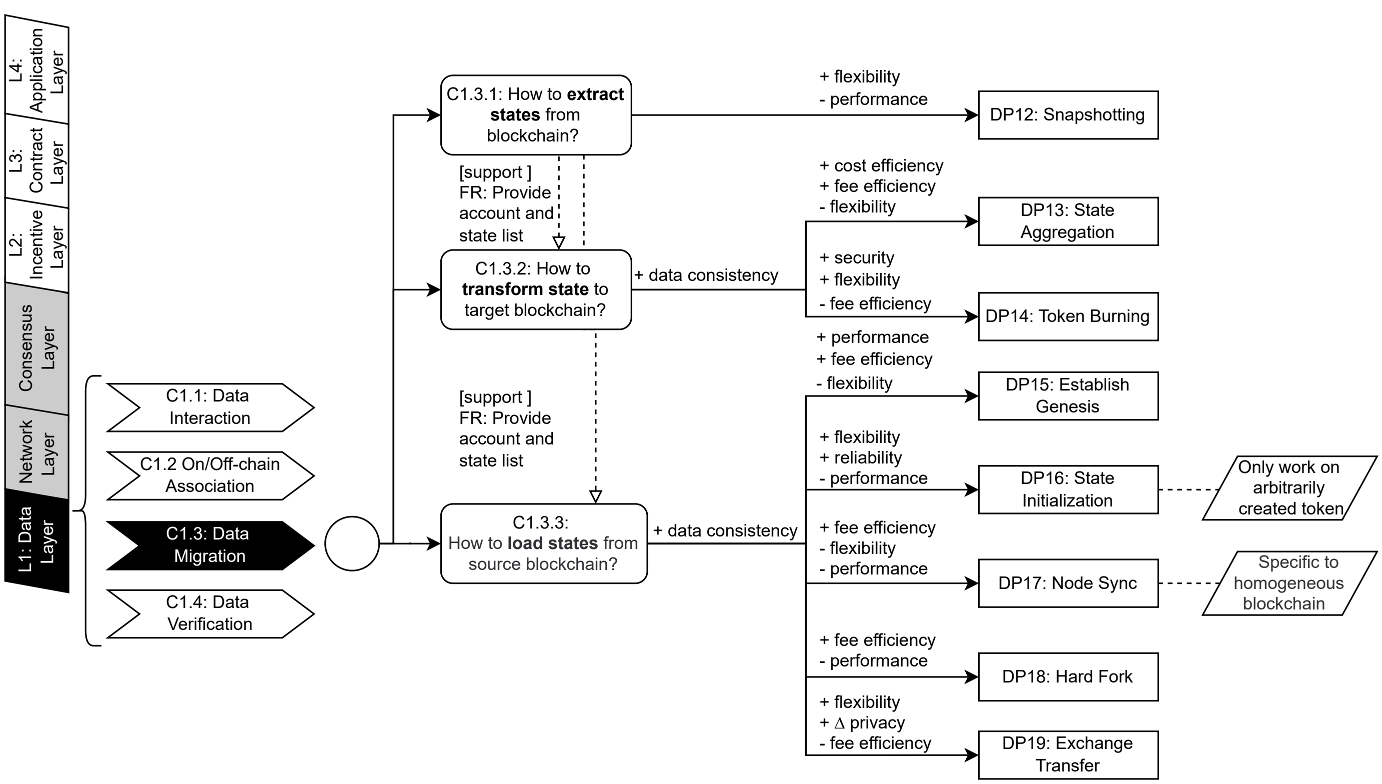

Fig. 6: Updated Version: Decision Model for Data Migration (C1.3)

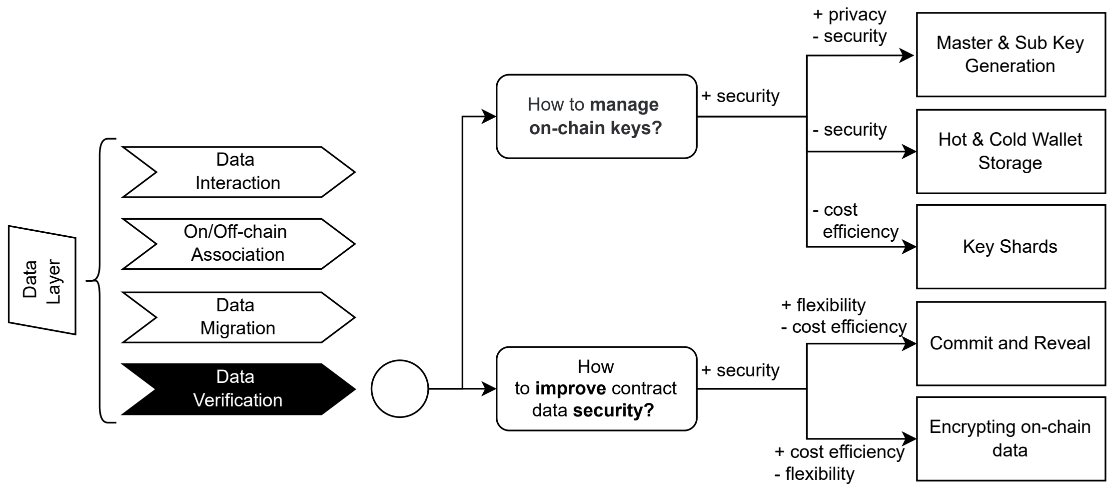

Fig. 7: Original Version: Decision Model for Data Verification (C1.4)

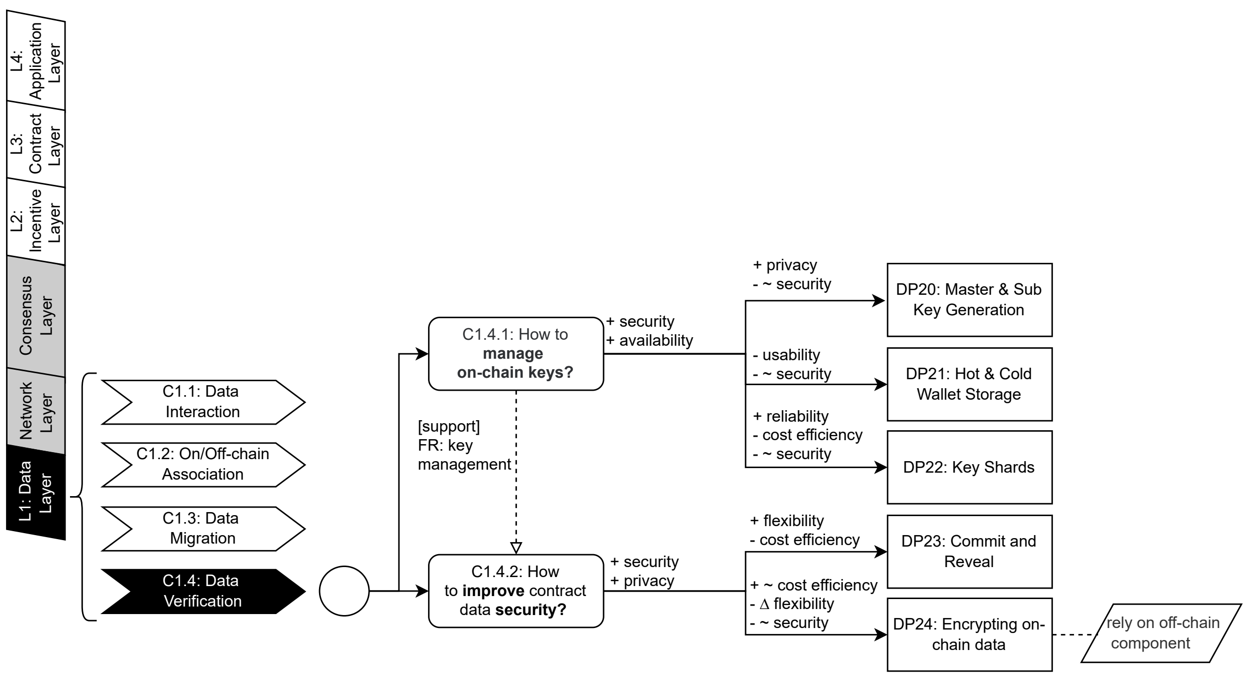

Fig. 8: Updated Version: Decision Model for Data Verification (C1.4)

to homogeneous blockchain were introduced to emphasize limitations and dependencies specific to certain design patterns.

2) **Integration of Blockchain Six-Layer Architecture:** The model now aligns with the six-layer blockchain architecture for better context and structure.
3) **Correction of Quality Attribute Mappings:** Quality attribute mappings have been adjusted for accuracy. For example, *Snapshotting* no longer incorrectly includes *data consistency*, and the negative impact on *performance* in *State Initialization* has been explicitly indicated.
4) **Inclusion of Flexibility and Other Missing Attributes:** The model now incorporates additional attributes, such as *flexibility*, ensuring a more comprehensive evaluation of design patterns and their trade-offs in practical scenarios.
5) **Item Numbering for Quick Reference:** All items, including challenges (*C1.3.1, C1.3.2, C1.3.3*) and patterns (*DP12, DP13, DP14, DP15, DP16, DP17, DP18, DP19*), have been systematically numbered to enhance accessibility and simplify navigation within the model.

C1.4: Data Verification

Updates to the Decision Model: The Data Verification decision model has been revised to include the relationships between typical challenges. Other elements such as quality attributes incorporation and annotation correction are also included in the updated model for better effectiveness. Below is a list of the major updates:

1) **Addition of Inherent Attributes:** Inherent attributes have been added to clarify dependencies. For example, *DP24: Encrypting on-chain data* now explicitly states that it *relies on off-chain components*.
2) **Integration of Blockchain Six-Layer Architecture:** The revised model aligns with the six-layer blockchain architecture.
3) **Defined Relationships Between Design Challenges:** Relationships between challenges have been explicitly defined. For instance, *C1.4.1* (*How to manage on-chain keys?*) supports *C1.4.2* (*How to improve contract data security?*) through providing key management.
4) **Enhanced Quality Attribute Scope:** Symbols like Δ and $\sim$ have been added to indicate trade-offs and impacts of quality attributes. Attributes such as *availability* and *privacy* are presented more clearly.
5) **Correction of Misleading Annotations:** Misleading annotations have been revised. For example, the security impact of *DP20: Master & Sub Key Generation* and *DP21: Hot & Cold Wallet Storage* has been corrected to more accurately reflect their trade-offs.
6) **Improved Quality Attribute Mapping:** Additional attributes such as *availability* and *privacy* have been incorporated, ensuring a more comprehensive evaluation of the design patterns.
7) **Item Numbering for Quick Reference:** Unique identifiers have been assigned to all items in the model, including challenges (*C1.4.1, C1.4.2*) and patterns (*DP20, DP21, DP22, DP23, DP24*), simplifying navigation and enabling quick reference.

L2: Incentive Layer

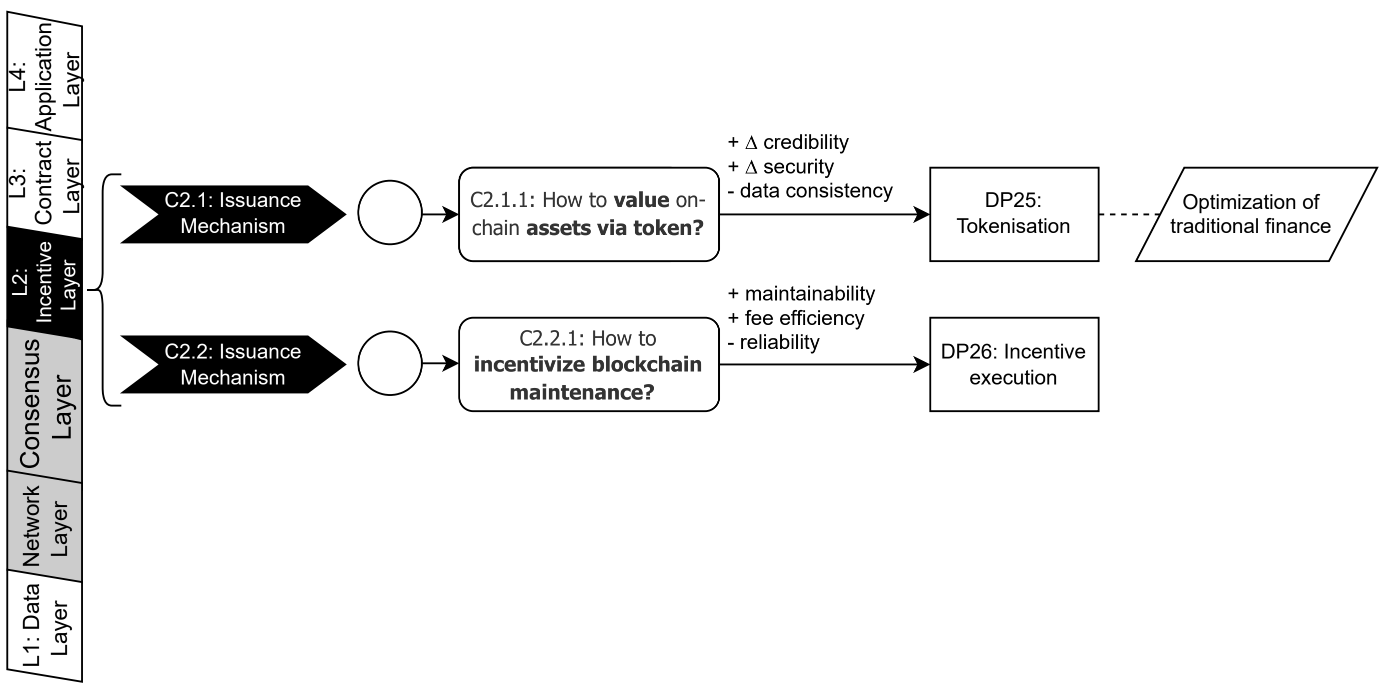

Fig. 9: Updated Version: Incentive Layer (L2) (Newly Extended)

Expansion to the Decision Model: Based on expert feedback, the decision model was expanded to include a new category: *L2: Incentive Layer*. This addition addresses an important layer of blockchain systems that focuses on incentivizing participation and maintaining the system's integrity. Below is a list of the updates:

1) **Introduction of Incentive Layer as a New Category:** A new layer, *L2: Incentive Layer*, was added to the model to address challenges related to token issuance and incentivizing blockchain maintenance. This addition ensures that the model reflects the economic mechanisms critical to blockchain systems.
2) **Inclusion of New Challenges:** Two new challenges were introduced under the Incentive Layer:
   - *C2.1.1: How to value on-chain assets via token?* This challenge focuses on mechanisms for assigning value to on-chain assets using tokenization.
   - *C2.2.1: How to incentivize blockchain maintenance?* This challenge addresses methods for encouraging participants to contribute to the maintenance of the blockchain network.
3) **Addition of Relevant Design Patterns and Quality Attribute Mapping Results:** Two design patterns were added to address the new challenges, including *DP25: Tokenisation* and *DP26: Incentive Execution*. The design patterns were mapped to relevant quality attributes to reflect their impact.

L3: Contract Layer

C3.1: Contract Authorization

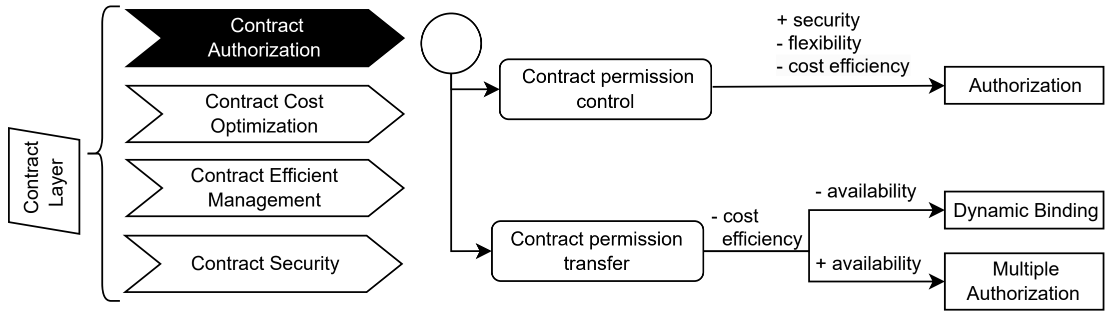

Fig. 10: Original Version: Decision Model for Contract Authorization (C3.1)

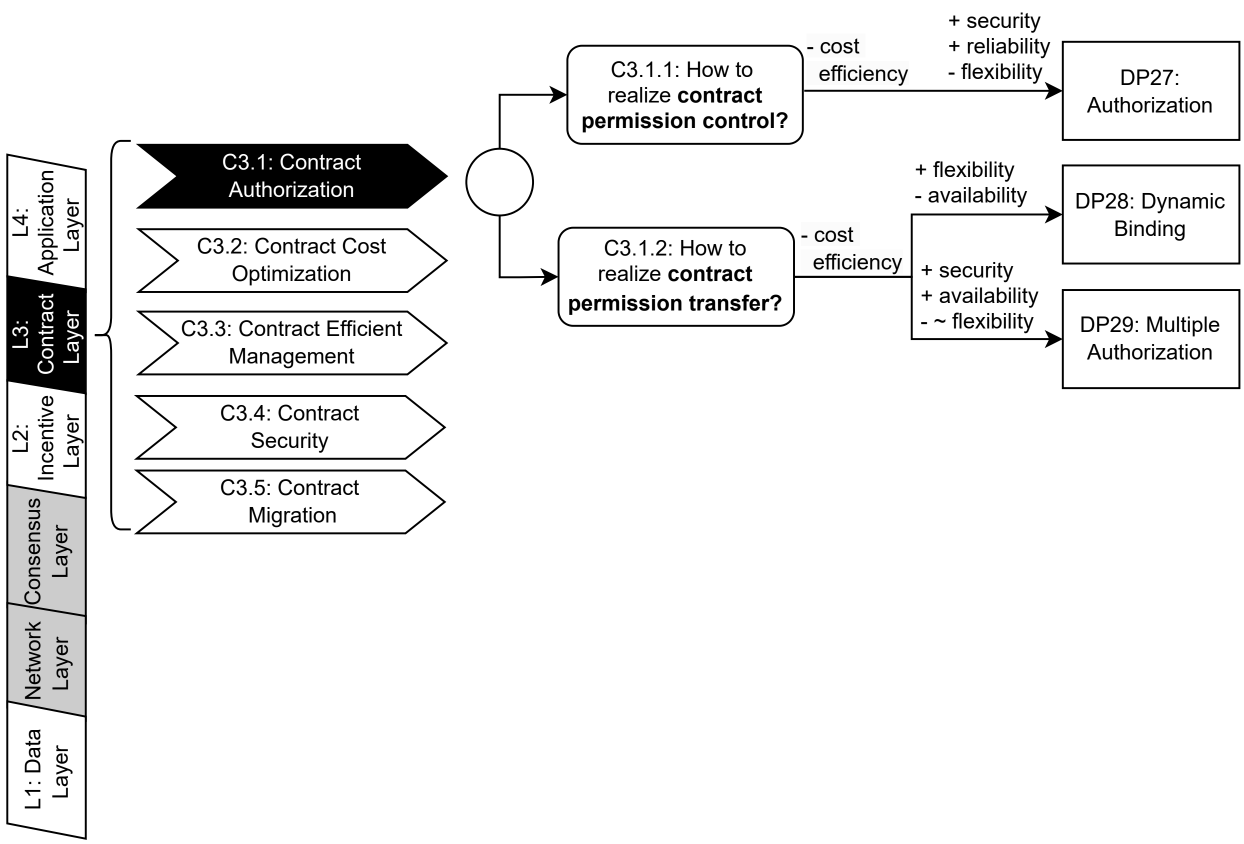

Fig. 11: Updated Version: Decision Model for Contract Authorization (C3.1)

Updates to the Decision Model: The Contract Authorization model was introduced to address challenges related to ensuring secure and reliable access control for smart contracts. Quality attribute mappings were refined to emphasize key attributes such as security, flexibility, and usability. Below is a list of the major updates:

1) **Integration of Blockchain Six-Layer Architecture:** The revised model now aligns with the six-layer blockchain architecture.
2) **Enhanced Quality Attribute Scope:** The updated model incorporates scope indicators such as $\sim$ to represent trade-offs and impacts of quality attributes. Attributes *flexibility* can be presented clearly.
3) **Improved Quality Attribute Mapping:** Additional attributes, such as *availability*, have been included and mapped to the corresponding design patterns, addressing previous oversights and ensuring a more comprehensive evaluation framework.
4) **Item Numbering for Quick Reference:** Each item in the model, including challenges (*C3.1.1*, *C3.1.2*) and patterns (*DP27*, *DP28*, *DP29*), has been assigned a unique identifier. This enhancement simplifies navigation and allows users to quickly locate specific elements within the decision model.

C3.2: Contract Cost Optimization

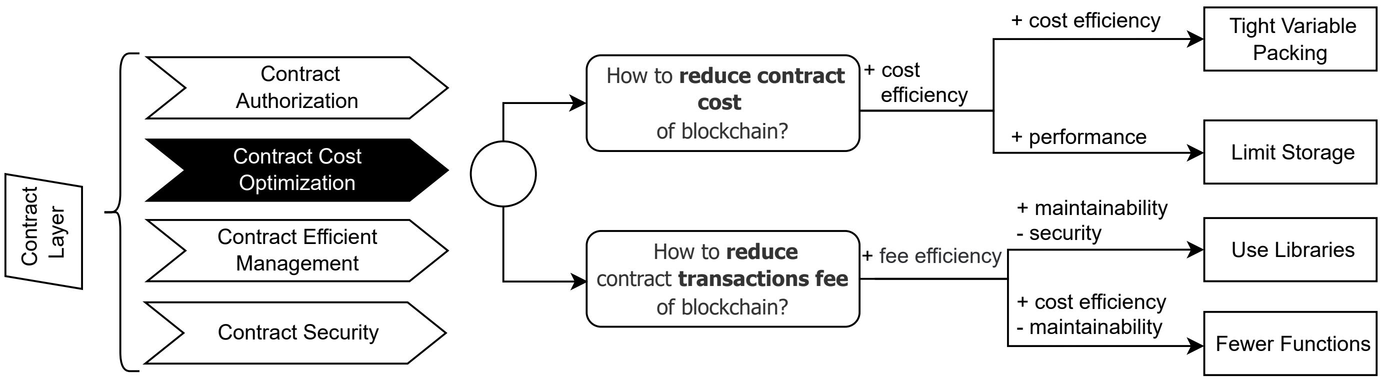

Fig. 12: Original Version: Decision Model for Contract Cost Optimization (C3.2)

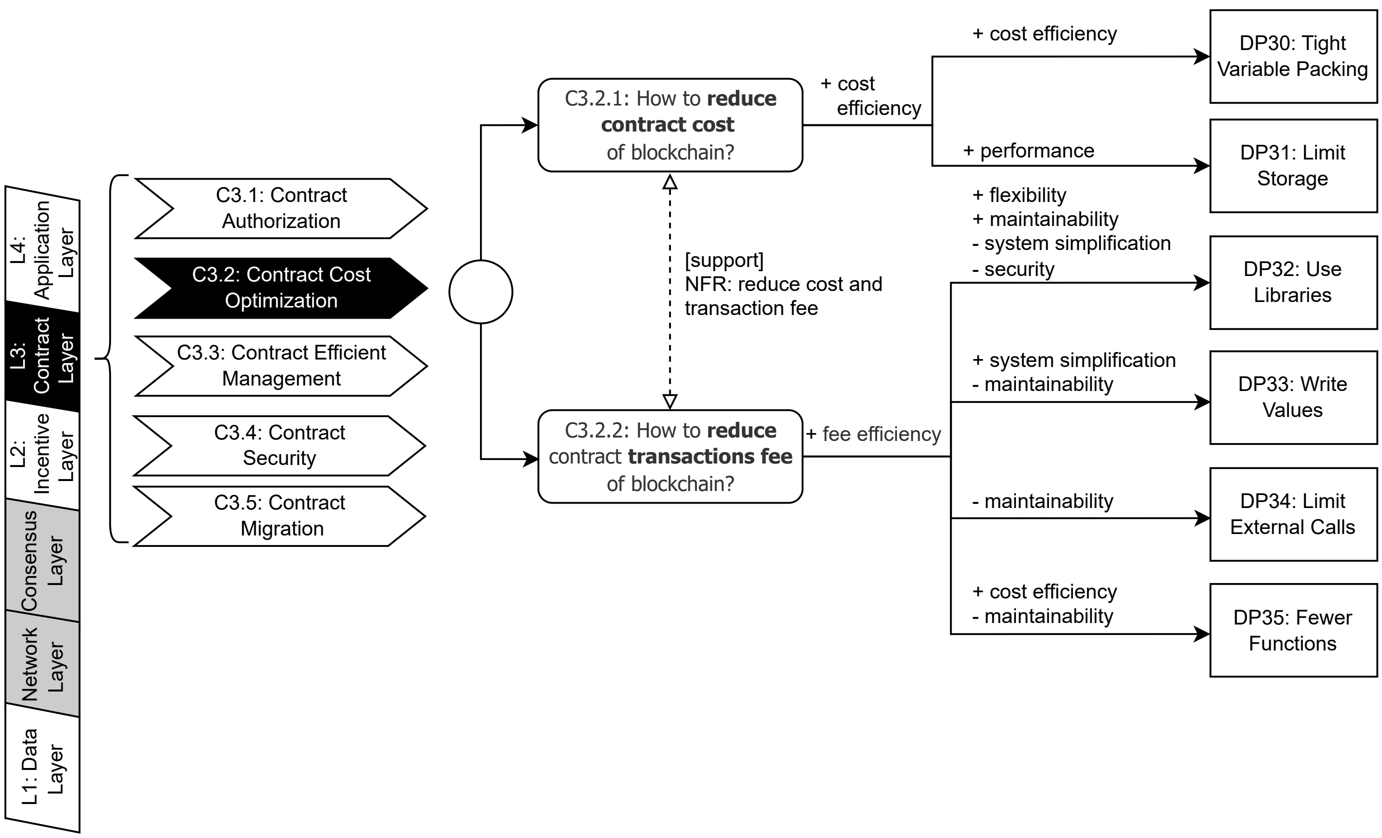

Fig. 13: Updated Version: Decision Model for Contract Cost Optimization (C3.2)

Updates to the Decision Model: The Contract Cost Optimization model was introduced to tackle challenges related to reducing the operational costs of deploying and executing smart contracts. New design patterns, such as Write Values and Limit External Calls, were added to support cost-efficient contract operations. Quality attribute mappings were refined to highlight key trade-offs. Below is a list of the major updates:

1) **Item Numbering for Quick Reference:** Unique identifiers have been added to all items in the model, including challenges (*C3.2.1*, *C3.2.2*) and patterns (*DP30*, *DP31*, *DP32*, *DP33*, *DP34*, *DP35*). This enables faster navigation and reference within the decision model.
2) **Integration of Blockchain Six-Layer Architecture:** The updated model aligns with the six-layer blockchain architecture, situating the *Contract Cost Optimization* layer in its appropriate context to provide a clearer understanding of its role within the overall framework.
3) **Defined Relationships Between Design Challenges:** The revised model explicitly identifies the bidirectional *[support]* relationship between *C3.2.1* (*How to reduce contract cost of blockchain?*) and *C3.2.2* (*How to reduce contract transaction fees of blockchain?*), highlighting how solutions to both challenges can reduce cost and transaction fee.

4) **Addition of Missing Design Patterns:** Missing patterns such as *DP33: Write Values* and *DP34: Limit External Calls* have been incorporated to ensure a comprehensive set of solutions.
5) **Improved Quality Attribute Mapping:** The updated model includes discussions on previously overlooked quality attributes such as *flexibility*, ensuring a more complete evaluation of trade-offs associated with the design patterns.

C3.3: Contract Management and Operation

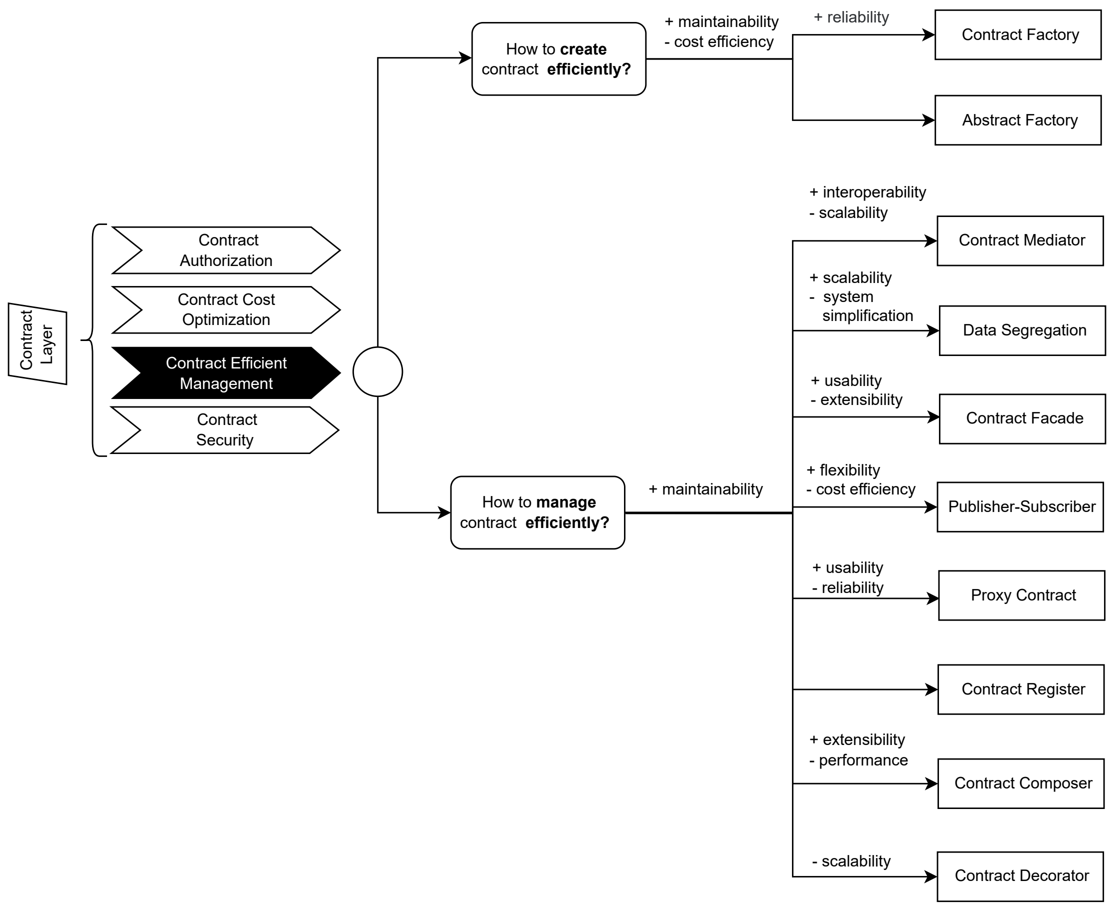

Fig. 14: Original Version: Decision Model for Contract Management and Operation (C3.3)

Updates to the Decision Model: The Contract Management and Operation model was introduced to address challenges related to the efficient deployment and updating of smart contracts. The typical challenges are reclassified and missing quality attributes are added such as system simplification and cost efficiency. Below is a list of the major updates:

1) **Item Numbering for Quick Reference:** Unique identifiers have been added to all items, including challenges (*C3.3.1*, *C3.3.2*, *C3.3.3*) and patterns (*DP36*, *DP37*, *DP38*, *DP39*, *DP40*, *DP41*, *DP42*, *DP43*, *DP44*, *DP45*, *DP46*, *DP47*). This ensures easier navigation and faster reference.
2) **Integration of Blockchain Six-Layer Architecture:** The model now aligns with the six-layer blockchain architecture, providing better context for the *Contract Efficient Management* layer's role within the broader blockchain framework.
3) **Reclassification of Challenges:** The challenge *How to manage contracts efficiently?* was too broad and has been split into more focused sub-challenges: "*C3.3.2: How to execute contracts efficiently?*" and "*C3.3.3: How to upgrade contracts efficiently?*". This reclassification enhances decision-making clarity.
4) **Addition of Missing Quality Attributes:** The updated model includes quality attributes such as *system simplification* and *cost efficiency*, which were previously omitted, ensuring a more thorough evaluation.
5) **Defined Relationships Between Design Patterns:** The model now describes the relationships between patterns. For instance, *DP40: Flyweight* is identified as a *sub-pattern* supporting *DP39: Data Segregation*.
6) **Inclusion of Missing Design Patterns:** Patterns such as *DP40: Flyweight* and *DP42: State Machine* have been added to fill gaps in the original model.
7) **Enhanced Quality Attribute Scope and Mapping:** Scope indicators like Δ and $\sim$ have been added, and more detailed mappings of positive and negative impacts on quality attributes have been incorporated. For example, the impact of *extensibility* and *performance* has been refined for each pattern, ensuring better differentiation among them.

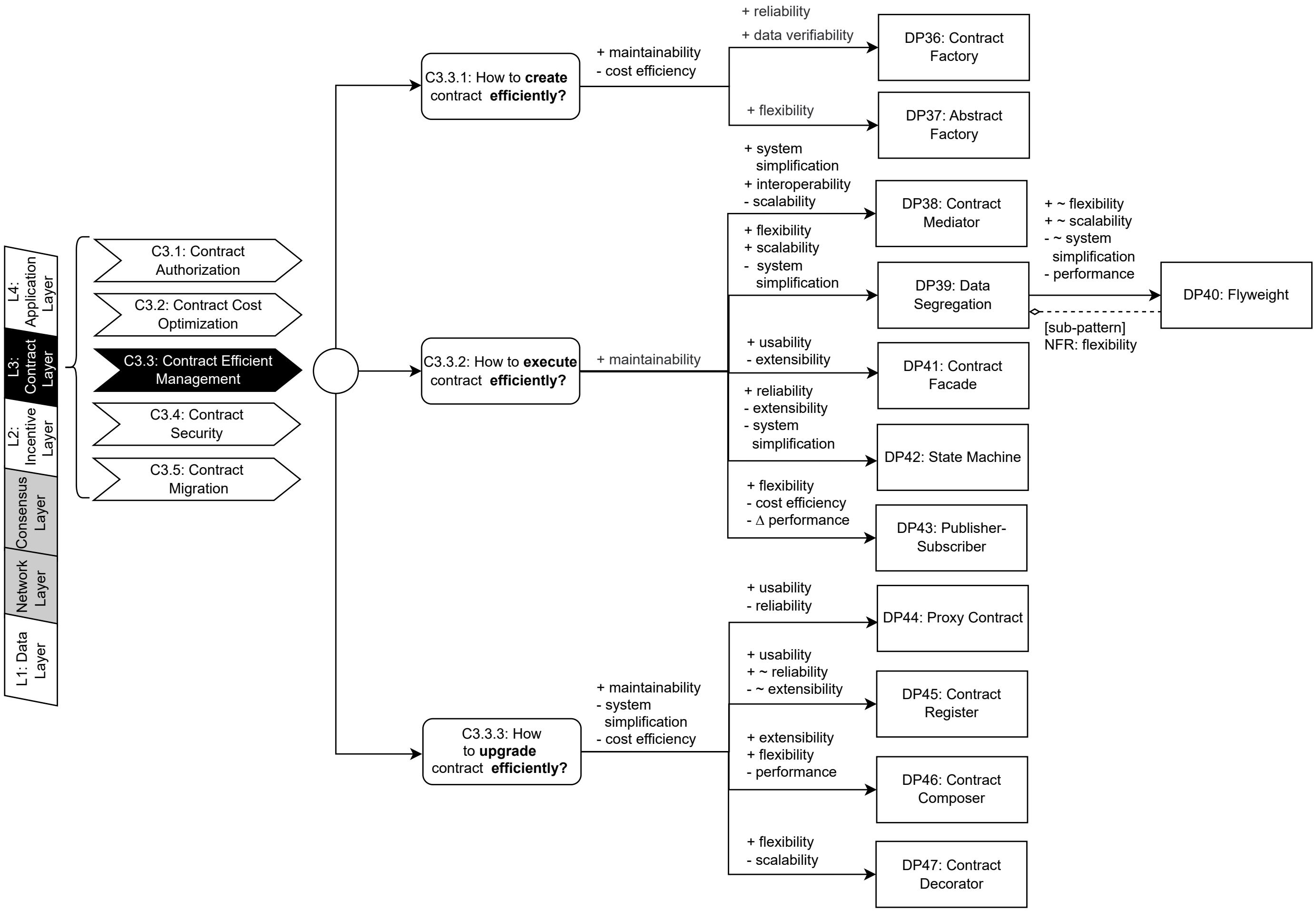

Fig. 15: Updated Version: Decision Model for Contract Management and Operation (C3.3)

C3.4: Contract Security

Updates to the Decision Model: The Contract Security model was introduced to ensure the safety against potential vulnerabilities and attacks. The relationship between challenges are defined and a new challenge has been added:"How to terminate contracts". Corresponding design patterns and quality attribute mappings are also introduced. Below is a list of the major updates:

1) **Addition of Missing Challenges and Design Patterns:** A new challenge, *C3.4.5: How to terminate contracts?*, has been added to address contract termination concerns. Corresponding design patterns, such as *DP57: Automatic Deprecation* and *DP58: Emergency Stop*, have also been incorporated.
2) **Defined Relationships Between Challenges:** Relationships between challenges have been identified, such as *C3.4.1* (*How to reduce the risk of attack on contracts?*), which is supported by *C3.4.4* (*How to prevent DoS attacks?*). These challenges address different attack types, highlighting their interconnections and complementary focus.
3) **Item Numbering for Quick Reference:** All items in the model have been assigned unique identifiers, including challenges (*C3.4.1, C3.4.2, C3.4.3, C3.4.4, C3.4.5*) and patterns (*DP48, DP49, DP50, DP51, DP52, DP53, DP54, DP55, DP56, DP57, DP58*). This allows users to locate elements quickly and efficiently.
4) **Integration of Blockchain Six-Layer Architecture:** The model now aligns with the six-layer blockchain architecture, situating the *Contract Security* layer in its appropriate structural context for better clarity.
5) **Inclusion of Inherent Attributes:** Inherent attributes such as *specific to latency* (for *DP55: Speed Bump*) and *specific to throughput* (for *DP56: Rate Limit*) have been added to distinguish between design patterns with similar functionality.
6) **Enhanced Quality Attribute Mapping:** Quality attribute mappings have been refined to ensure a more comprehensive evaluation. For example, the impacts on *flexibility*, *reusability*, and *cost efficiency* are now explicitly detailed for relevant patterns.

C3.5: Contract Migration

Extension to the Decision Model: The Contract Migration model was newly introduced to address the emerging need for transferring smart contracts across different blockchains or platforms. This addition reflects the growing importance of operability in blockchain systems. Below is a list of the major updates:

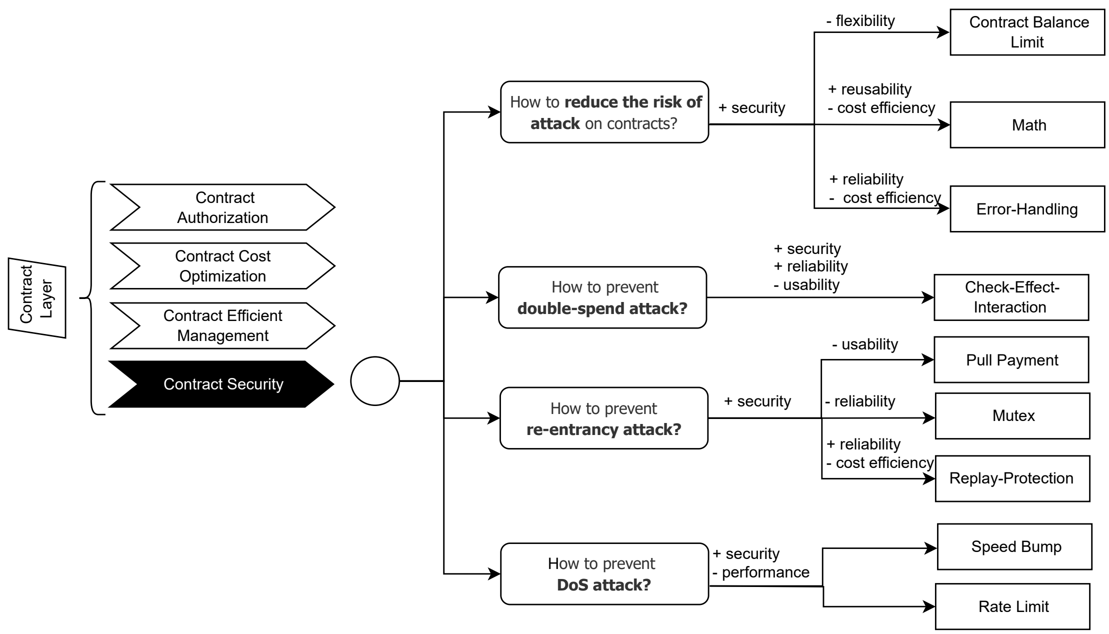

Fig. 16: Original Version: Decision Model for Contract Security (C3.4)

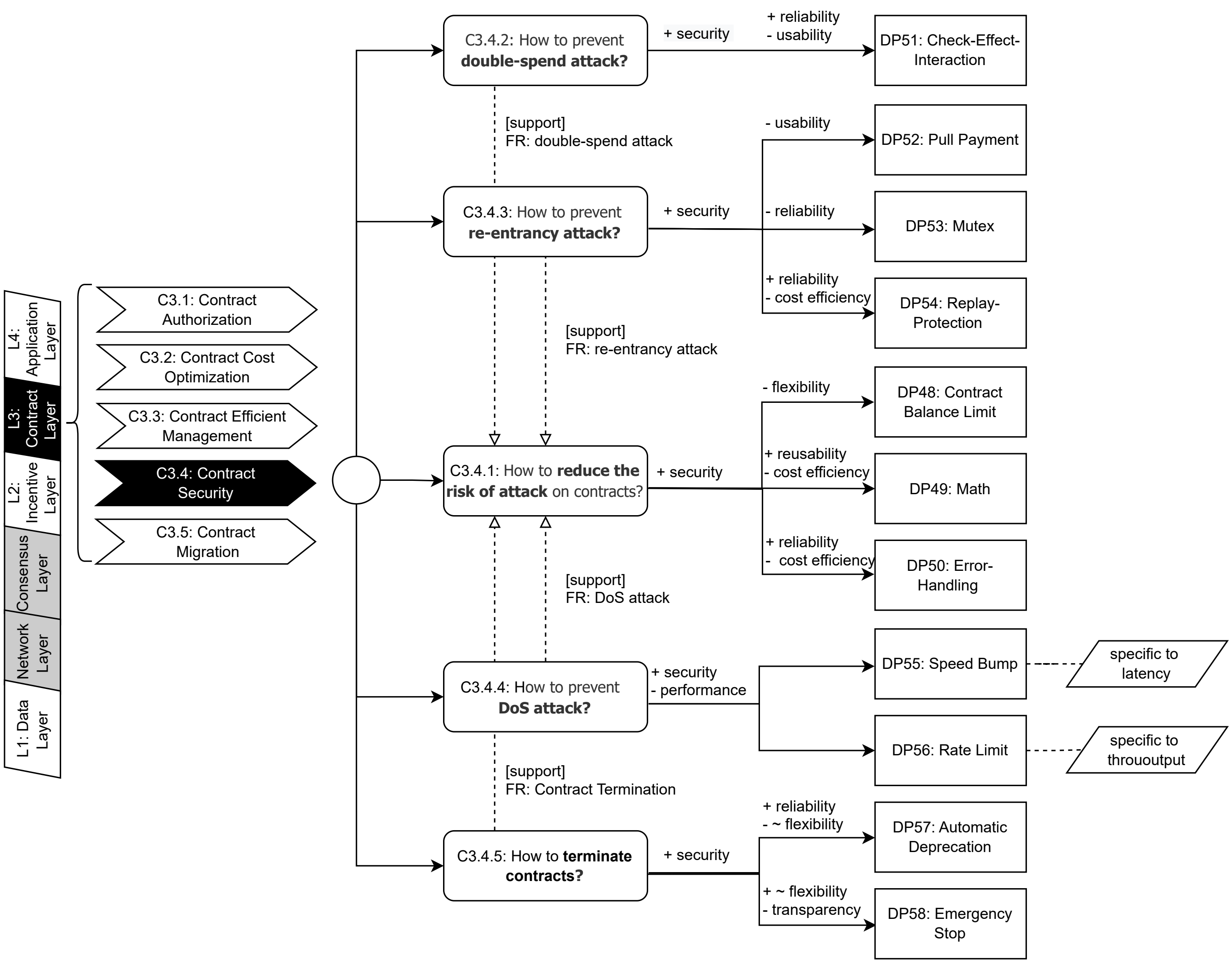

Fig. 17: Updated Version: Decision Model for Contract Security (C3.4)

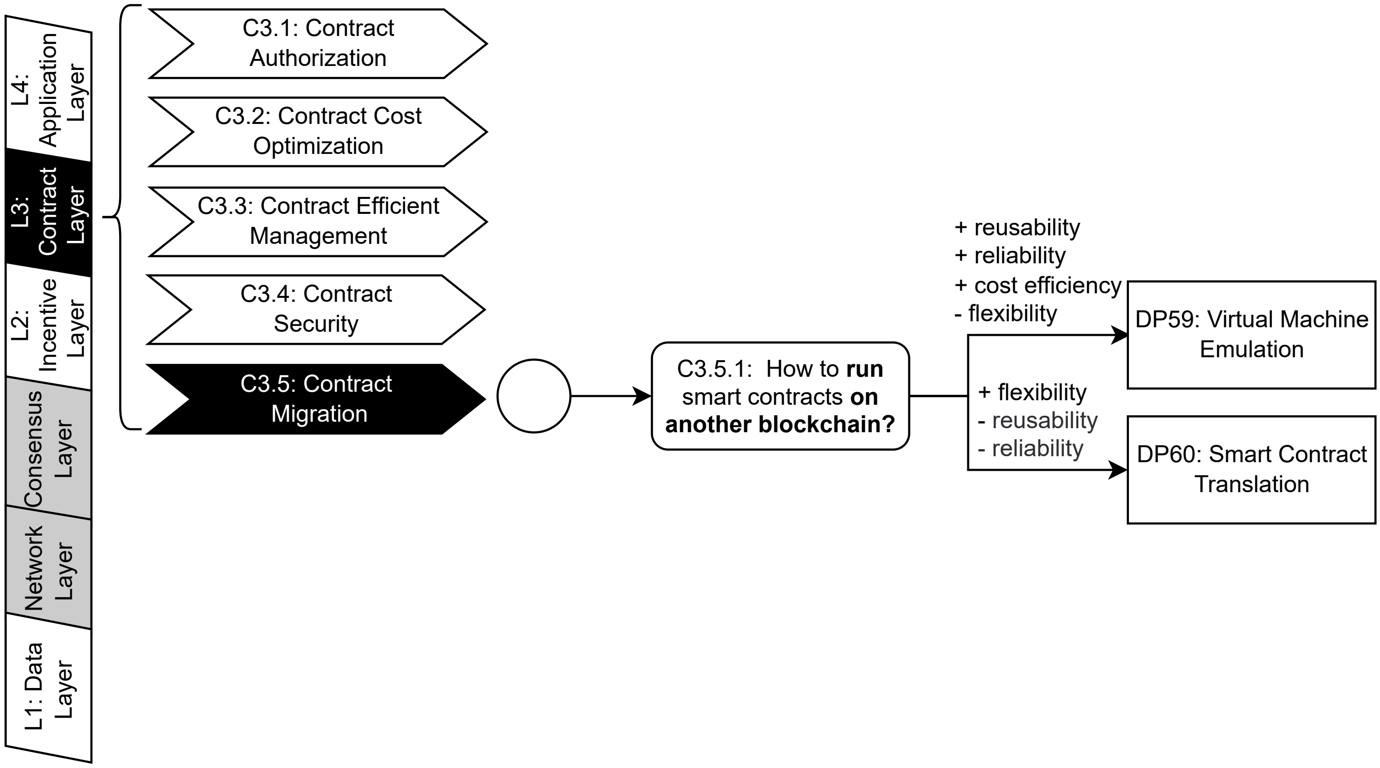

Fig. 18: Updated Version: Decision Model for Contract Migration (C3.5) (Newly Extended)

1) **Introduction of Contract Migration as a New Category:** A new category, *C3.5: Contract Migration*, was added to the decision model to address challenges related to running smart contracts across different blockchains. This ensures comprehensive coverage of contract-related issues.
2) **Inclusion of a New Challenge:** The challenge *C3.5.1: How to run smart contracts on another blockchain?* was introduced to capture the core issue of cross-chain contract migration. This challenge highlights the importance of interoperability and flexibility in blockchain systems.
3) **Addition of Relevant Design Patterns and quality attribute mappings:** Two design patterns, *DP59: Virtual Machine Emulation* and *DP60: Smart Contract Translation*, were added under the new challenge. These patterns provide solutions for enabling smart contracts to operate on different blockchains, addressing key quality attributes such as: reusability, reliability, etc.

L4: Application Layer

C4.1: Identity Management and Verification

Updates to the Decision Model: The Identity Management model was expanded to include verification, aligned with the blockchain six-layer architecture, and updated with unique identifiers for easier reference. Missing design patterns, such as One-Off Access, were added, inherent attributes introduced for context, and quality attribute mappings refined to better address trade-offs like privacy and cost. Below is a list of the major updates:

1) **Item Numbering for Quick Reference:** Each component of the model, including challenges (e.g., *C4.1.1*, *C4.1.2*), design patterns (e.g., *DP61*, *DP62*, *DP63*), and quality attributes, has been assigned a unique identifier. This allows users to quickly locate and reference specific elements within the model.
2) **Incorporation of the Blockchain Six-Layer Architecture:** The revised model integrates the blockchain six-layer architecture, providing a clearer context for decision-making.
3) **Updated name of Identity Management:** The original name of the concern: *Identity Management* has been updated to include *Verification*, presenting both secure identity management and user identity validation challenges.
4) **Enhanced Representation of Quality Attribute Scopes:** The updated model uses symbols such as Δ and $\sim$ to explicitly represent the scope and trade-offs of quality attributes. This provides a clearer understanding of the quality attributes like *security* and *data verifiability*.
5) **Introduction of Inherent Attributes:** Inherent attributes, such as *optimization of traditional identity management* and *optimization of media account management*, have been added. These provide additional context for understanding the implications of specific design patterns.
6) **Addition of Missing Design Patterns:** Missing design patterns, such as *One-Off Access*, were incorporated into the updated model. The relationships between this and other patterns: *Time-Constrained Access*, are now represented.

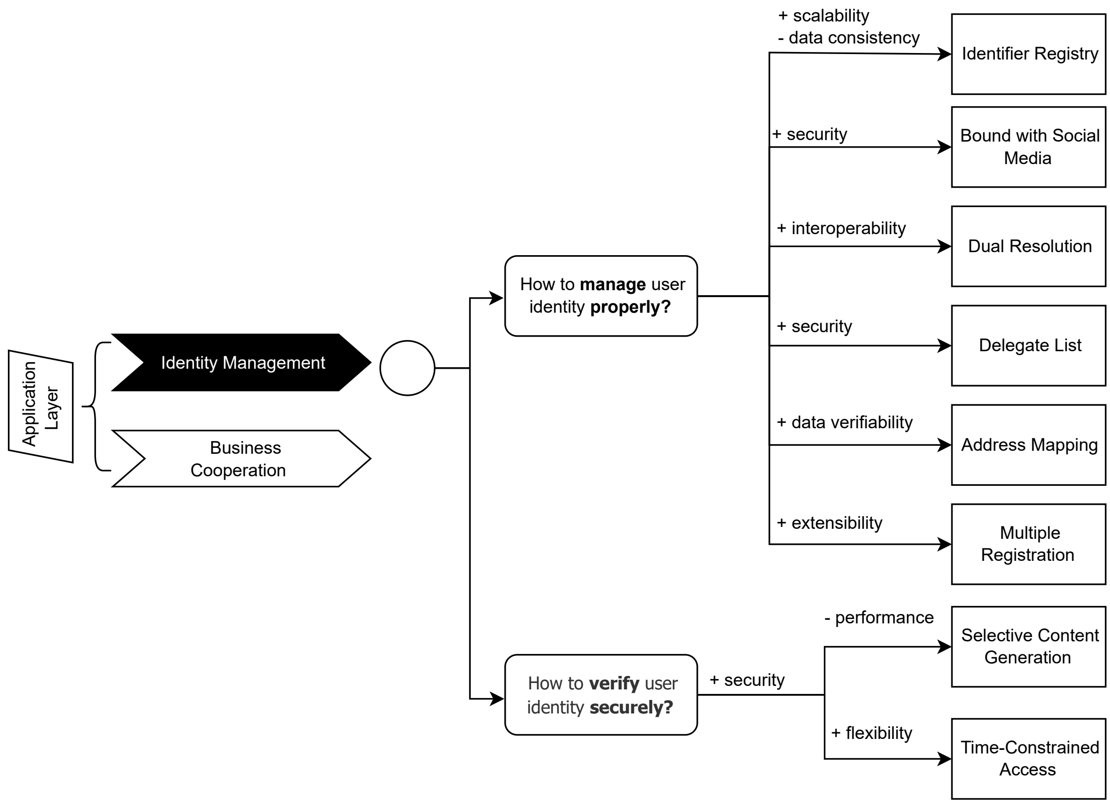

Fig. 19: Original Version: Decision Model for Identity Management and Verification (C4.1)

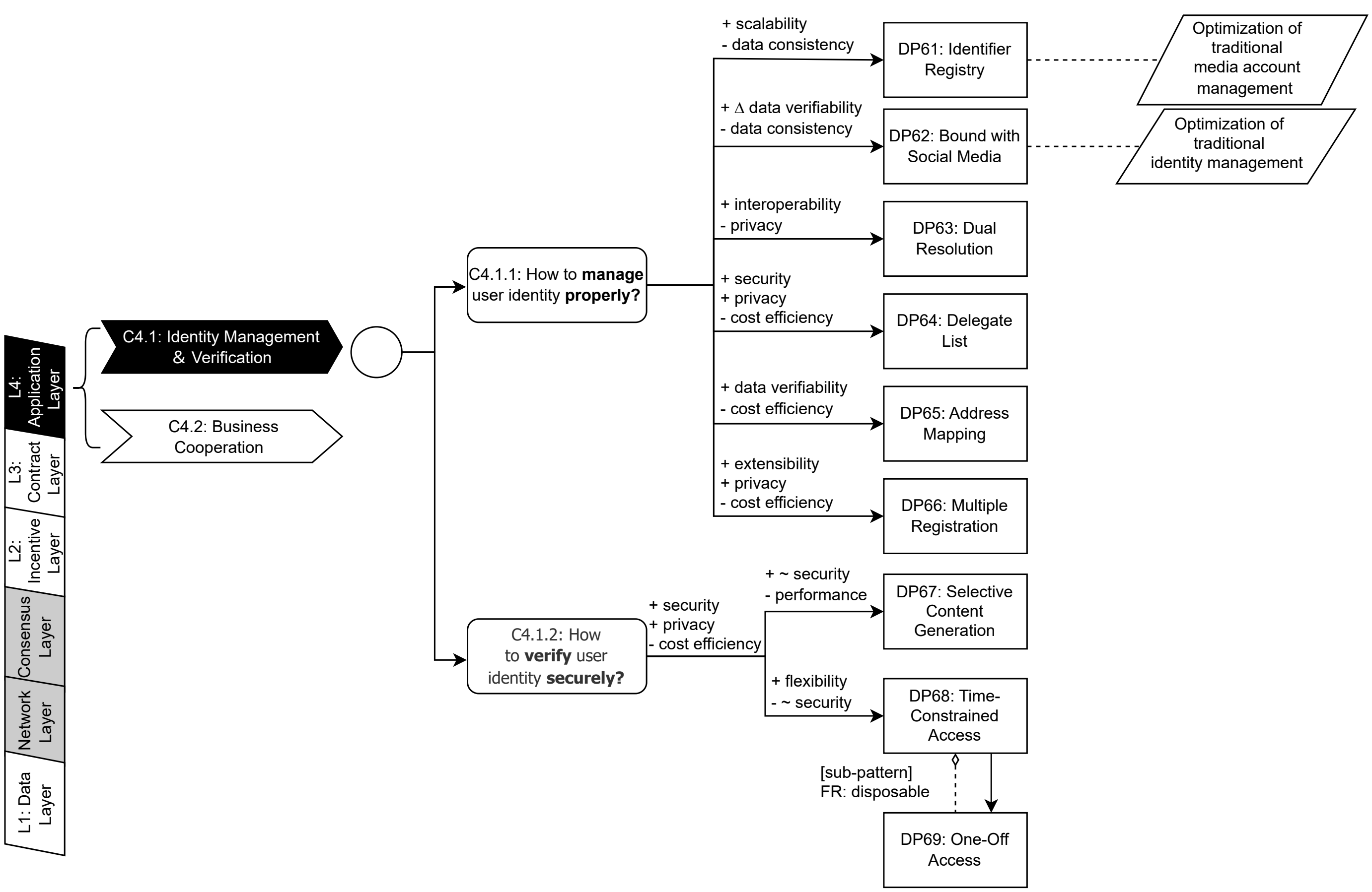

Fig. 20: Updated Version: Decision Model for Identity Management and Verification (C4.1)

7) **Improved Quality Attribute Mapping:** The revised model expands the discussion of key quality attributes, such as *privacy* and *cost efficiency*. It also corrects previous mapping errors, such as the inaccurate mapping between *Bound with Social Media* and *security*, ensuring that each pattern is correctly aligned with its relevant attributes.

C4.2: Business Cooperation

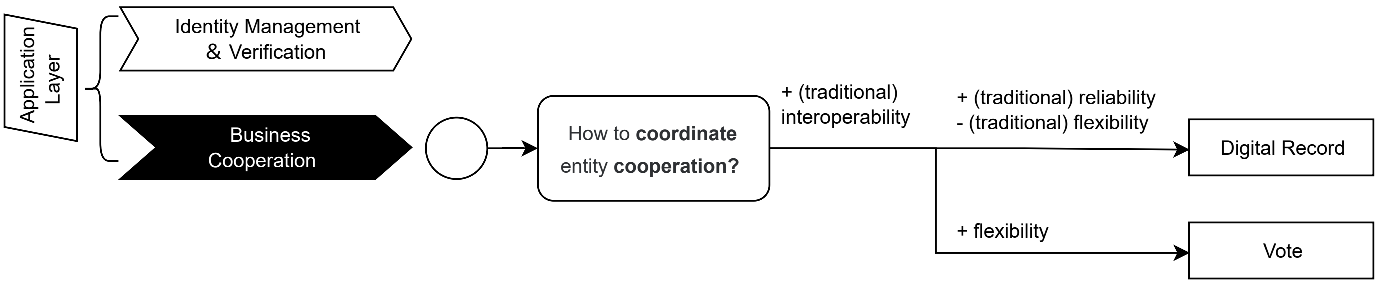

Fig. 21: Original Version: Decision Model for Business Cooperation (C4.2)

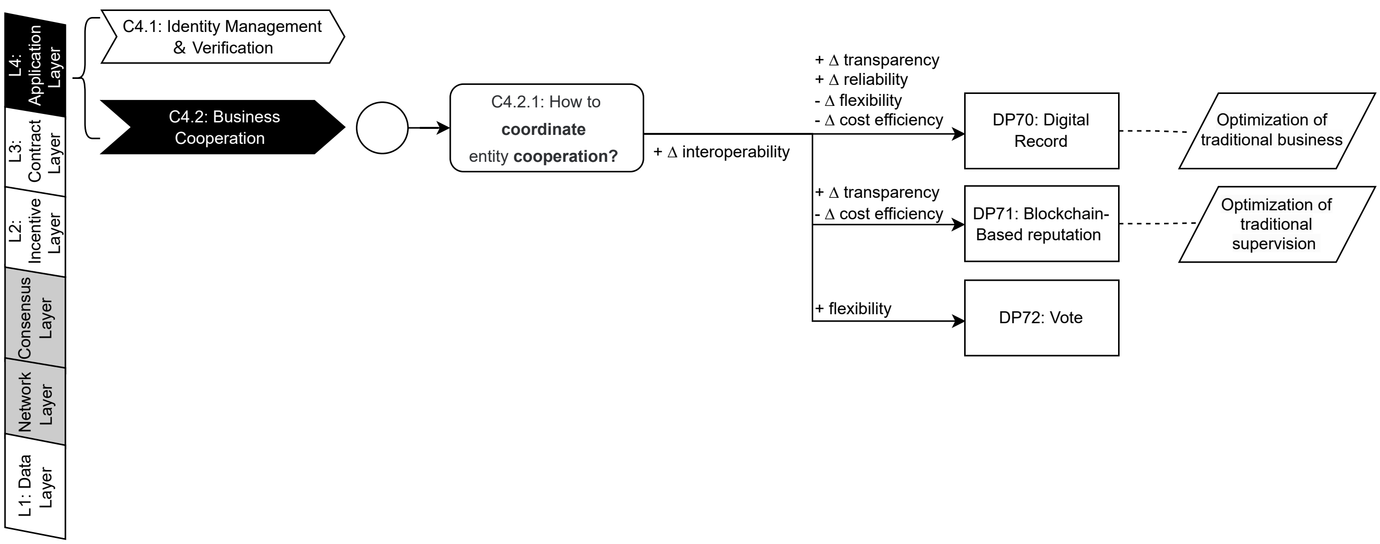

Fig. 22: Updated Version: Decision Model for Business Cooperation (C4.2)

Updates to the Decision Model: The Business Cooperation model was updated with the addition of missing design patterns like Blockchain-Based Reputation. Inherent attributes, such as Optimization of traditional business, were introduced to clarify pattern distinctions, while quality attribute mappings were refined to better address key factors like cost efficiency and transparency. Below is a list of the major updates:

1) **Item Numbering for Quick Reference:** Each component of the model, including challenges (*C4.2.1*) and design patterns (*DP70*, *DP71*, *DP72*), has been assigned a unique identifier. This enables users to quickly locate and reference specific items within the model.
2) **Integration of Blockchain Six-Layer Architecture:** The revised model incorporates the blockchain six-layer architecture, aligning challenges and design patterns with their corresponding layers.
3) **Enhanced Scope Representation of Quality Attributes:** Initial discussions of quality attribute scopes, marked as "(traditional)," have been refined using symbols such as Δ *transparency*, Δ *reliability*, and Δ *flexibility*.
4) **Incorporation of Missing Design Patterns:** The missing design pattern *Blockchain-Based Reputation* has been added to the model, ensuring a more complete representation of decision-making options.
5) **Introduction of Inherent Attributes:** Inherent attributes, such as *Optimization of traditional business* and *Optimization of traditional supervision*, have been added to provide additional context and clarify distinctions between similar design patterns like *Digital Record* and *Blockchain-Based Reputation*.
6) **Improved Quality Attribute Mapping** The revised model expands the discussion of quality attributes such as *cost efficiency* and *transparency*, ensuring a more robust evaluation of design patterns. This improves the model's ability to guide effective decision-making.